CHECKLIST OF COMPLIANCE WITH THE SYLLABUS PARTS AND FORMAT

Course Code and Title**: S-CSSE321 SOFTWARE ENGINEERING 1 LECTURE**

This undertaking has a two-fold objective. First, it allows the syllabus designers to review their output systematically. Based on their honest evaluation, they could make the necessary revision. Second, it expedites the checking to be done by the Dept. Syllabus Committee Reviewer, Department Chair, the College Dean, and the Institutional Syllabus Reviewers.

DIRECTIONS: Review the latest version of the syllabus that you submitted to the Department Chair for his/her scrutiny. As you do your reflective review, assess your compliance with the syllabus parts and format by answering each of the items that follow. Shade the corresponding box (complied with, not complied with) under the compliance column.

Format, parts, and description	Compliance	
	Complied With	Not Complied With
Format		
1. The 2023-2024 syllabus format is used.	/	
2. The font used in the syllabus preparation is Times New Roman (11).	/	
3. DLSUD (being first) and college logos are beside each other at the top (center).	/	
4. Preliminaries include all the contact details of the professor, such as the day and time.	/	
5. Each WEEK has a corresponding date.	/	
Content		
6. The Learning Outcomes have more HOTS than LOTS.	/	
7. CLOs and TLOs are identified for each module/lesson/topic.	/	
8. Gospel reading and Reflection precede the presentation of each topic under the teaching-learning activities.	/	
9. The journal articles/references are within the last five years.	/	
10. Some materials are beyond the five-year rule because these are considered cannon and seminal studies.	/	
11. The course policies and requirements section of the syllabus is a clear provision for addressing the concerns and grievances of students.	/	
Assessments		
12. At least three and a maximum of five Enabling Assessments per term are reflected in the syllabus.	/	
13. Formative Assessments are not reflected anymore in the syllabus.	/	
14. Enabling Assessments are briefly described and aligned with the learning outcomes and teaching-learning activities.	/	
15. The syllabus presents one-module(lesson)-one-assessment correspondence.	/	
16. Only one Enabling Assessment is reflected in a given week	/	
17. The Enabling Assessments are aligned with the identified CLOs and TLOs.	/	
18. Enabling Assessments, Class Participation, and Summative Assessments constitute the breakdown of grades per term.	/	

19. Thirty percent (30%) is allotted for Summative Assessments per term.	/	
20. Twenty percent (20%) is allotted for Class Participation/Attendance per term.	/	
21. The percentage that is allocated for Enabling Assessments fairly corresponds to their quantity, depth, breadth, and level of difficulty compared to the other components of the Midterm Grade and Final Grade.	/	
22. Summative Assessments are aligned with the identified CLOs, TLOs and teaching-learning activities.	/	
23. Enabling Assessments are aligned with the identified CLOs and TLOs and teaching-learning activities.	/	
24. The Midterm Summative Assessment measures appropriately and validly with the identified CLOs and TLOs for the entire MIDTERM PERIOD	/	
25. The Final Summative Assessment measures appropriately and validly with the identified CLOs and TLOs for the entire FINAL PERIOD.	/	
References and Curriculum Map		
26. The References must be within the last 5 years except for canons and seminal studies.	/	
27. It must be a combination of books, eBooks, journal articles, videos, etc.	/	
28. It should follow the APA 7th edition for CLAC, CBAA, CTHM, CCJE, CEAT, COED, and CSE for CSCS.	/	
29. The syllabus is submitted with an attached mapping table (CLOs with PLOs and ILOs)	/	

NOTE: Enabling Assessments include Discussion Type (cf. SB), Forum, Analysis Paper, Reports, etc.

Self-reviewed:



Sheryl D. Kamantigue
Faculty

Checked, verified, and endorsed:

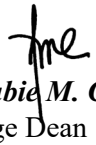


Rolando B. Barrameda
Dept. Curriculum Committee Reviewer

Checked, verified, and approved:



Josephine T. Eduardo
Department Chair



Dr. Rubie M. Causaren
College Dean

DE LA SALLE UNIVERSITY-DASMARINAS
COLLEGE OF SCIENCE AND COMPUTER STUDIES
COMPUTER STUDIES DEPARTMENT

COURSE SYLLABUS

COURSE CODE	**:**	S-CSSE321
COURSE TITLE	**:**	Software Engineering 1
COURSE TYPE	**:**	Lecture
COURSE CREDIT	**:**	2 Units
PRE-REQUISITES	**:**	S-ITCS226LA
CO-REQUISITES	**:**	None
MODALITY	**:**	Traditional Blended Learning
PROFESSOR	**:**	SHERYL D. KAMANTIGUE
CONTACT DETAILS:		sdkamantigue@dlsud.edu.ph / MS Teams / SB – M-F, 8-5PM

COURSE DESCRIPTION:

This course deals with equipping the students with the skills in developing quality software which covers topics such as concepts of software engineering, use of software engineering paradigms as applied to software development, planning and managing software projects, software development life cycle and various software processing models. Various case studies will be given, and students are expected to develop quality software.

The course will have a special provision on modality, pacing, pedagogy, assessment type, communication and feedback based on the national, regional, provincial, city, and institutional guidelines for the safety amidst disaster and pandemic. Hence, this class shall be delivered in both online and offline/module-based approach. This course is also compliant with Outcomes-Based Education (OBE) and 21st Century Learning Design (21CLD) frameworks.

COURSE LEARNING OUTCOMES:

By the end of this course, students can:

CLO 1. understand software engineering concepts by identifying the characteristics, application, life cycle of software and design principles and techniques;
CLO 2. apply the most appropriate software process model;
CLO 3. analyze agile software development methods;
CLO 4. evaluate software requirements and appraise the processes involved in discovering and documenting these requirements;
CLO 5. examine different system models that may be developed as part of requirements engineering and system design processes;
CLO 6. create research by developing quality software, applying the concepts learned, and write a technical paper about it; and
CLO 7. demonstrate 21st century skills in all learning activities.

LEARNING PLAN:

Learning Outcomes for Midterm Period

Course Learning Outcomes	Topic Learning Outcomes
CLO 1. Understand software engineering concepts by identifying the characteristics, application, life cycle of software and design principles and techniques.	• TLO 1. Describe what software engineering is and why it is important. • TLO 2. Recognize that the development of different types of software system may require different software engineering techniques. • TLO 3. Identify ethical and professional issues that are important for software engineers. • TLO 4. Describe the four systems, of different types, which are used as examples throughout the book.
CLO 2. Apply the most appropriate software process model.	• TLO 5. Define the concepts of software processes and software process models. • TLO 6. Discuss the three general software process models and when they might be used. • TLO 7. Explain the fundamental process activities of software requirements engineering, software development, testing, and evolution. • TLO 8. Recognize why processes should be organized to cope with changes in the software requirements and design. • TLO 9. Identify the notion of software process improvement and the factors that affect software process quality.
CLO 3. Analyze agile software development methods.	• TLO 10. Compare the rationale for agile software development methods, the agile manifesto, and the differences between agile and plan-driven development. • TLO 11. Distinguish about important agile development practices such as user stories, refactoring, pair programming and test-first development. • TLO 12. Relate the Scrum approach to agile project management. • TLO 13. Organize the issues of scaling agile development methods and combining agile approaches with plan-driven approaches in the development of large software systems.
CLO 4. Evaluate software requirements and appraise the processes involved in discovering and documenting these requirements.	• TLO 14. Evaluate the concepts of user and system requirements and why these requirements should be written in different ways. • TLO 15. Value the differences between functional and nonfunctional software requirements. • TLO 16. Appraise the main requirements engineering activities of elicitation, analysis, and validation, and the relationships between these activities. • TLO 17. Weigh why requirements management is necessary and how it supports other requirements engineering activities.

CLO 7. Demonstrate 21st century skills in all learning activities.	• *Collaboration, Skilled Communication, Knowledge Construction, Self-Regulation, Real-World Problem-Solving, Use of ICT (not part of topic, but integrated in assessments)*

Module No and Learning Outcome(s)	**Date**	**Teaching-Learning Activities / Assessment Strategies** **Online Module Activities**	**Technology Enabler**	**Asynchronous**	**Onsite**	**Allotted Hours**
Module 1 TLO 1 TLO 2 TLO 3 TLO 4 CLO 1 CLO 7	August 14-19	Gospel reading, prayer and reflection Syllabus discussion Presentation of course introduction Project Grouping (3 members per group) Discussion of Project Specification Discussion of Module 1 – Introduction Discuss Professional Software Development	Schoolbook MS Teams PowerPoint		/	2
	August 21-26	Gospel reading, prayer and reflection Continuation of Module 1 – Introduction (Self-paced reading) Discuss Software Engineering Ethics Discuss Four Sample Case Studies	MS Teams PowerPoint	/		1
		EA: Online Short Quiz on Module 1	Schoolbook	/		1

Module 2 TLO 5 TLO 6 TLO 7 TLO 8 TLO 9 CLO 2 CLO 7	August 28-September 2	Gospel reading, prayer and reflection Discussion of Module 2 – Software Processes Discuss Software process models Discuss Process activities Discuss Coping with change Discuss Process improvement	MS Teams PowerPoint		/	1
		Gospel reading, prayer and reflection Self-paced reading of Module 3 – Agile Software Development Agile methods Agile development techniques	Schoolbook		/	1
	September 4-9	Gospel reading, prayer and reflection Continuation of Self-paced reading of Module 3 – Agile Software Development Agile project management Scaling agile methods		/		2
Module 3 Module 4 TLO 10 TLO 11 TLO 12	September 11-16	Gospel reading, prayer and reflection EA: Seatwork: At the end of your study program, students in a software engineering course are typically expected to complete a major project. Explain how	MS Teams PowerPoint		/	1

TLO 13 TLO 14 TLO15 TLO 16 TLO 17 CLO 3 CLO 4 CLO 7		the agile methodology may be very useful for the students to use in this case.				
		Gospel reading, prayer and reflection Discussion of Module 3 – Agile Software Development Agile methods Agile development techniques Agile project management Scaling agile methods	Schoolbook		/	1
	September 18-23	Gospel reading, prayer and reflection Self-paced Reading of Module 4 – Requirements Engineering	Schoolbook PowerPoint	/		2
	September 25-30	Gospel reading, prayer and reflection EA: Online Submission of Chapter 1 and Chapter 2 of the Software Development Project	MS Teams PowerPoint	/		2
	October 2-7	Gospel reading, prayer and reflection Discussion of Module 4 – Requirements Engineering	Schoolbook		/	2
	October 9-14	MIDTERM EXAMINATION			/	2
			TOTAL	***6***	***12***	***18***

Learning Outcomes for Final Period

Course Learning Outcomes	Topic Learning Outcomes
CLO 5. Examine different system models that may be developed as part of requirements engineering and system design processes.	• TLO 18. Relate how graphical models can be used to represent software systems and why several types of model are needed to fully represent a system. • TLO 19. Analyze the fundamental system modeling perspectives of context, interaction, structure, and behavior. • TLO 20. Distinguish the principal diagram types in the Unified Modeling Language (UML) and how these diagrams may be used in system modeling. • TLO 21. Relate to model-driven engineering, where an executable system is automatically generated from structural and behavioral models.
CLO 6. Create research by developing quality software, applying the concepts learned, and write a technical paper about it.	• TLO 22. Develop a fully running system as a group project, together with a documentation and defend it in class.
CLO 7. Demonstrate 21st century skills in all learning activities.	• *Collaboration, Skilled Communication, Knowledge Construction, Self-Regulation, Real-World Problem-Solving, Use of ICT (not part of topic, but integrated in assessments)*

Module No and Learning Outcome(s)	Date	Teaching-Learning Activities / Assessment Strategies	Technology Enabler	Asynchronous	Onsite	Allotted Hours
Module 5 TLO 18 TLO 19 TLO 20 TLO 21 TLO 22 CLO 5 CLO 6 CLO 7		**Online Module Activities**				
	October 16-21	(STUDENT'S WEEK) Gospel reading, prayer and reflection Discussion of Module 5 – System Modeling	Schoolbook PowerPoint	/		1
		EA: Assignment: Develop a Use Case Diagram based on your group project.	PowerPoint Schoolbook	/		1
	October 23-28	SELF CARE WEEK				2
	October 30-November 4	Gospel reading, prayer and reflection Continuation of Module 5 - System Modeling EA: Quiz on Module 5	Schoolbook PowerPoint	/		2
	November 6-11	Gospel reading, prayer and reflection Project Development Time	Schoolbook MS Teams	/		2
CLO 6 CLO 7	November 13-18	Gospel reading, prayer and reflection Project Consultation	MS Teams Schoolbook		/	1
		Gospel reading, prayer and reflection EA: Initial Submission of Project (Software and Documentation)	Schoolbook		/	1
	November 27-December 2	Gospel reading, prayer and reflection EA: Final Submission of Project (Software and Documentation)	Schoolbook		/	2
	December 4-9	Gospel reading, prayer and reflection Summative Assessment: Project Defense / Presentation	PowerPoint		/	2
	December 11-16	Gospel reading, prayer and reflection Continuation of Summative Assessment: Project Defense / Presentation	PowerPoint		/	2
			TOTAL	***6.0***	***10.0***	***16.0***

GRADING SYSTEM:

Midterm

Class Participation / Attendance	- 20%
Enabling Assessments	- 50%
Summative Assessments	- 30%
	100%

Finals

Class Participation / Attendance	- 20%
Enabling Assessments	- 50%
Summative Assessments	- 30%
	100%

Overall Final Grade = (Midterm + Final) / 2

COURSE POLICIES AND REQUIREMENTS:

1. **Office365 Activation.** Please ensure that your Office365 account is working. Your Office365 account is needed to access both Schoolbook and MS Teams where your asynchronous and synchronous classes will be held.
2. **Enrollment in an E-Class.** You will automatically be enrolled in your e-class which is based on your enrollment data.
3. **Enriched Virtual Blended Learning Model.** This course adopts the enriched virtual blended model. This means that there will be a mix of face-to-face, synchronous, and asynchronous activities. The majority of teaching-learning activities and assessments are undertaken online. The total number of onsite classes shall be 30% of the number of hours allotted for the whole semester.
4. **Online Asynchronous Sessions**.
    a. **Schoolbook (SB).** Schoolbooks shall be the only platform for asynchronous sessions.
    b. **Modules.** Modules are self-paced learning resources for asynchronous sessions. These can be accessed in Schoolbook.
    c. **References.** Each page section may contain uploaded references. These learning resources may be downloaded.
    d. **Asynchronous Activities.** You are expected to read the modules as soon as they are uploaded. The learning content of the modules complements the online synchronous and face-to-face sessions.
    e. **Asynchronous Engagement.** Your activities in the course can be tracked by your professor. This includes the time you spend in reading the lessons and answering the assessments.
    f. **Schoolbook Forum.** All general concerns about the lessons and assessments in asynchronous sessions must be posted in the Schoolbook Forum. A response shall be made by your teacher within 48 hours.
    g. **Schoolbook Messaging.** This shall be the mode of communication for private and/or confidential communications. A response shall be made by your teacher within 48 hours upon receipt of the same unless it falls on weekends or holidays, which shall be handled promptly the following working day.
5. **Onsite / Face-to-face (F2F) Sessions.**
    a. **Face-to-face engagement.** Your engagement in face-to-face classes is graded based on your class participation.

b. **Classroom.** F2F classes shall be held in the classroom indicated in your Certificate of Registration. Should there be changes in the classroom venue, information will be given in advance.
c. **Gospel Reading and Prayer.** Each F2F session shall start with a Gospel reading and prayer. Your teacher may assign you, in advance, to do this.
d. **F2F Meeting Schedule.** The meeting schedule shall follow the time indicated in your official registration. The dates of F2F meetings are identified in the learning plan.
e. **Attendance.** Attendance in F2F meetings is required.
f. **Tardiness.** A student who came 1 minute to 15 minutes after the start of the face-to-face meeting is considered late. Three tardy attendance is equivalent to 1 absence.
g. **Absence.** A student is considered absent 15 minutes after the official class schedule.
h. **Excuse F2F classes.** Students are excused in the F2F classes based on the provisions in the latest version of the Student Handbook.
i. **Uniform.** Wearing of uniform is optional. Civilian attire should follow the policy on dress code as stipulated in the latest version of the Student Handbook.

6. **Assessment and Grading System.**

a. **Formative assessments.** These are ungraded assessments. These may be considered practice assessments that lead toward achieving outcomes without fear of receiving a failing grade.
b. **Enabling assessments.** These will comprise most of your graded assessments. These are designed to achieve topic learning outcomes, that lead to achieving the course learning outcomes. Only one enabling assessment shall be allowed during the week. Please pay attention to the duration and number of attempts. As a general rule, quiz-type enabling assessments shall be open for only a minimum of 24 hours, while output-based enabling assessments shall be open for at least 6 days.
c. **No. of Attempts.** All enabling assessments, whether given onsite or online, shall have a maximum of 3 attempts only. Summative assessments shall be given a maximum of 2 attempts.
d. **Summative assessments.**
    i. There shall be two summative assessments (midterm and final exams) for the entire semester. These are designed to achieve the course learning outcomes.
    ii. Online summative assessments shall be uploaded on SB and shall be accessible for a minimum of 1 day for quiz type and a minimum of 6 days for output-based assessments.
    iii. Summative assessment may be given online or face-to-face.
    iv. Output-based summative assessment shall be given to students at least fifteen days prior to the scheduled Summative Exam Week.
e. **Lifeline.** Only students with (1) valid reason as stated in the Student Handbook and IRR, and (2) given their proof of excuse on or before the next synchronous/F2F session, shall be given a lifeline on the enabling and summative assessments.
f. **Rubric.** All online non-quiz or non-discrete types of assessment (essay, dropbox, output-based, etc.) shall have a rubric or criteria for rating the students' tasks. A student may refuse to answer these types of assessments in the absence of a rubric or criteria for grading, and the assessment shall be deemed invalid and shall not be part of the student's grades.
g. **Grading.** All online assessments should be checked and graded by the teacher before the submission of midterm and final grades.
h. **Grading system.** The following shall be the basis for the computation of grades per term for fully online and enriched virtual blended classes (see table above):
    i. Enabling Assessments: 50%
    ii. Class Participation (including Attendance): 20%
    iii. Summative Assessments: 30%
i. **Gradebook.** Students can see the breakdown of grades in their Assessment tab.

7. **Self-Care.**

   a. **Schedule.** There is only one schedule of Self-Care week during the Second Semester SY2022-2023. It is on April 24 to 29. During this week, there shall be no synchronous meetings, F2F classes, new modules, new assessments, and deadlines.
   b. **Prerogative.** Students may avail of the self-care program, whether online or onsite, provided by the different units of the University.

8. **Copyright and Plagiarism.**

   a. Using of any illegally obtained software and other technology tool is strictly prohibited.
   b. Students are encouraged to use their original photos, videos, and other resources. Otherwise, students can use royalty-free resources or embed the sources in their submissions to avoid copyright infringement and/or plagiarism.
   c. Giving of password to Schoolbook and Office 365 is strictly prohibited. Likewise, accessing Schoolbook and Office 365 account other than the students' personal account is also strictly prohibited. Violating students will be reported to the Student Welfare and Formation Office (SWAFO).
   d. This subject shall abide by the policies pertaining to intellectual property, copyright, and plagiarism as stipulated in the latest edition of the Student Handbook.
   e. Any plagiarized work, whether in part or full, shall mean a grade of 0.0 for the assessment.

9. This course shall abide by any institutional policies that may be released after the approval of this syllabus. Any such policy shall be posted within the e-class at the forums section, news feed. It will also be briefly discussed during the soonest synchronous meeting.

REFERENCES:

Online References:

Accession number or e-provider	**Reference Material**
e-Book (downloaded)	Sommerville, Ian. (2017). Software Engineering. 10th Global Edition. Pearson Education Ltd.
Web Resource	Alliance Software. (2020). An Introduction to Software Development Methodologies. Retrieved from https://www.alliancesoftware.com.au/introduction-software-development-methodologies/
Web Resource	Muhammad Raza. (2020). The Software Development Life Cycle: An Introduction. Retrieved from https://www.bmc.com/blogs/sdlc-software-development-lifecycle/
Web Resource	Carriou Erwan. (2019). Designing System. Retrieved from https://medium.com/swlh/designing-systemsce28ee31c0cd
Web Resource	Study.com (2021). Video Team: Requirements Modelling in Software Engineering. Retrieved from https://study.com/academy/lesson/requirements-modeling-in-software-engineering-classes-functionsbehaviors.html
Web Resource	Guru99. (2020) What is System Testing? Types and Definition. Retrieved from https://www.guru99.com/system-testing.html

On-Site References

Provider	**References**
AEA Lib	Leach, Ronald J. (2016). Introduction to Software Engineering. 2nd Edition. Boca Raton. CRC Press, Taylor & Francis Group

AEA Lib	Pressman, Roger S. (2015). Software engineering: a practitioner's approach. Singapore: McGraw-Hill Book Co.

Prepared: **Computer Studies Department**
1st Semester 2023-2024



Approved: **Josephine T. Eduardo**
Chair, CSD



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Dean, CSCS

De La Salle University - Dasmariñas
COLLEGE OF SCIENCE AND COMPUTER STUDIES
DEPARTMENT

PROGRAM LEARNING OUTCOME - COURSE LEARNING OUTCOME MAPPING TABLE FOR BS COMPUTER SCIENCE

COURSE CODE : S-CSSC321
COURSE TITLE : Software Engineering 1
COURSE TYPE : Lecture
COURSE CREDIT : 2 Units

Course Learning Outcome	Program Learning Outcome							
	PLO 1	**PLO 2**	**PLO 3**	**PLO 4**	**PLO 5**	**PLO 6**	**PLO 7**	**PLO 8**
1. Understand software engineering concepts by identifying the characteristics, application, life cycle of software and design principles and techniques.	X							X
2. Apply the most appropriate software model.	X							
3. Analyze agile software development methods.	X				X		X	
4. Evaluate software requirements and appraise the processes involved in discovering and documenting these requirements.	X	X						
5. Examine different system models that may be developed as part of requirements engineering and system design process.	X		X		X			
6. Create research by developing quality software, applying the concepts learned and write a technical paper about it.	X	X	X		X		X	X
7. Demonstrate 21st century skills in all learning activities	X	X	X		X			

NOTE: Provide a check mark on the areas in which the program learning outcome (PLO) is hit by the course learning outcome (CLO)

De La Salle University - Dasmariñas

COLLEGE OF SCIENCE AND COMPUTER STUDIES
COMPUTER SCIENCE DEPARTMENT

COMMON GE LEARNING OUTCOME - COURSE LEARNING OUTCOME MAPPING TABLE FOR BS COMPUTER SCIENCE

COURSE CODE: S-CSSC321
COURSE TITLE: Software Engineering 1
COURSE TYPE : Lecture
COURSE CREDIT: 2 Units

Course Learning Outcome	Common GE Learning Outcome				
	CGE1	CGE2	CGE3	CGE4	CGE5
1. Understand software engineering concepts by identifying the characteristics, application, life cycle of software and design principles and techniques.	X		X	X	
2. Apply the most appropriate software model.			X	X	
3. Analyze agile software development methods.		X	X	X	
4. Evaluate software requirements and appraise the processes involved in discovering and documenting these requirements.			X	X	
5. Examine different system models that may be developed as part of requirements engineering and system design process.			X	X	
6. Create research by developing quality software, applying the concepts learned and write a technical paper about it.		X	X		
7. Demonstrate 21st century skills in all learning activities		X	X		

NOTE: Provide a check mark on the areas in which the program learning outcome (PLO) is hit by the course learning outcome (CLO)

COMMON GE LEARNING OUTCOME

1. Critiques/evaluates individual's character, actions, and decisions based on Christian-like perspective
2. Assesses situations and proposes ways that benefit the country, particularly its environment and its marginalized sector
3. Explains in various occasions the relationship of the course/subject with the other subjects within or outside the discipline
4. Recommends ways to improve the subject/course

De La Salle University - Dasmariñas
COLLEGE OF SCIENCE AND COMPUTER STUDIES
COMPUTER SCIENCE DEPARTMENT

GRADUATE ATTRIBUTES (DESCRIPTORS/INSTITUTIONAL LEARNING OUTCOMES) PROGRAM LEARNING OUTCOME MAPPING TABLE FOR BS COMPUTER SCIENCE

Program Learning Outcome	Graduate Attributes/Institutional Learning Outcomes				
	GA1	GA2	GA3	GA4	GA5
1. Apply concepts of computing and mathematics in different domains of computer science.				X	X
2. Identify and analyze problems and define computing requirements appropriate to their solutions.					X
3. Design, develop, test and evaluate various solutions using algorithms for complex computing problems that take full advantage of usability of computer resources.			X	X	X
4. Document computer-based systems by writing effective reports for risk- assessment impact.				X	X
5. Work effectively as an individual or as a member of a team during software development.			X	X	X
6. Communicate effectively with the computing community and non-computer science professionals about computing activities.				X	X
7. Recognize the legal, social, ethical and professional issues involved in the utilization of computer technology and address these issues.	X	X	X		
8. Engage in continuing professional development through performing independent or collaborative research as computing professionals.			X	X	X

NOTE: Provide a check mark on the areas in which the Graduate Attribute (Descriptors/Institutional Learning Outcome) is hit by the program learning outcome (PLO). Kindly refer to the descriptors (institutional learning outcomes) to clearly understand what each attribute refers to or expect from its graduates.

Graduate Attribute (GA)	**(Institutional Learning Outcome)[1]**
1. God-centered	a) Creates an environment where the experience of God is lived and shared b) Practices honesty, fairness, truth and integrity in all aspects of life (personal and professional lives) c) Observes and maintains ethical standards in dealing with the different stakeholders d) Integrates Christian perspectives and values in all undertakings e) Manifests humility and respect in relating with other people
2. Patriotic	a) Manifests a deep sense of nationalism by integrating history, arts, and culture in their daily lives b) Participates responsibly and collaboratively in the discussion and resolution of issues within local, national, and international contexts c) Engages actively in political, social, economic, and cultural transformation for nation building d) Brings pride and honor to the community and the country e) Patronizes locally produced products and promotes them globally
3. Socially Responsive	a) Demonstrates concern and compassion for the plight of the vulnerable and marginalized sectors of society by participating meaningfully in the process of social transformation b) Engages in ecological advocacies as responsible stewards of God's creation c) Continuously works in solidarity with people and institutions to effect liberating action d) Brings forth awareness and deeper understanding of social realities e) Manifests spirit of volunteerism through social actions that improve communities
4. Globally Competitive	a) Takes progressive responsibility for own learning and development b) Promotes critical and creative thinking, self-knowledge and self-mastery c) Is committed to rise above mediocrity d) Shows dedication, commitment and love for work to attain exceptional results e) Exhibits innovativeness and creativity in various contexts
5. Professionally Competent	a) Possesses leadership qualities of **C**ompetence, **H**onesty, **A**ccountability, **R**esponsibility and **T**ransparency (CHART) b) Effectively communicates in various forms of media c) Strives to be at the top of their chosen fields d) Effectively works and collaborates in a multi-disciplinary team e) Adapts to changes in the workplace, local society and global community